9C Lesson 6

Ecology: abundance and distribution

Year 9 Science · 9C Threat from disease · Student Book pages 48 to 49

Name _____ Class _____ Date _____

Today I can:

☐ I can choose suitable apparatus for measuring the abundance and distribution of organisms.

☐ I can use data from a sample to estimate a population size and say how accurate the estimate is.

Key words

abundance how common a species is in a habitat
distribution how organisms are spread out in an area
population all the organisms of one species in an area
sample a small selected part of the total number
estimate an approximate calculation of something
quadrat a square frame used to sample plants or slow-moving animals
transect a line along which samples are taken
pitfall trap a container sunk into the ground to catch small ground animals
sweep net a net swept through long grass to catch insects
random chosen by chance, not by the person
biodiversity the number of different species in an area

● Core Everyone does this section.

1. Match each word to its definition. Draw a line.

abundance	an approximate calculation of something
distribution	the number of different species in an area
sample	how common a species is in a habitat
estimate	a small selected part of the total number
biodiversity	how organisms are spread out in an area

2. Tick one box in each row to show the best method for sampling each organism.

Organism	pitfall trap	sweep net	quadrat	pooter
ground beetles				
insects in long grass				
daisies on a lawn				
tiny insects on a leaf				
limpets on a rocky shore				

3. Students counted one species of snail in eight quadrats placed at random in a field 30 m by 25 m. Each quadrat was 0.5 m by 0.5 m.

Quadrat	1	2	3	4	5	6	7	8
Number of snails	3	1	0	2	4	1	2	3

a) Calculate the total number of snails counted.

b) Calculate the total area sampled by the eight quadrats, in m2.

c) Calculate the area of the whole field, in m2.

d) Estimate the population of this snail species in the whole field. Show your working.

4. Record your own quadrat results, then estimate the population of your plant in the 10 m by 10 m area.

a) Complete the table.

Quadrat	Coordinates	Number of plants
1		
2		
3		
4		
5		
6		
7		
8		
total		

b) Estimate the population: total x 100 m2 / 2 m2. Show your working.

●● Stretch When Core is finished.

S1. Give one reason for using quadrats to estimate a population size rather than counting all the snails.

S2. Suggest the best way to decide where to place the eight quadrats in the field. Explain your answer.

S3. Snails of another species in the field gather in groups during the day. Explain how this could affect an estimate of their population size made with the same method.

●●● Challenge Explain in full sentences.

C1. Ecologists compared the biodiversity of birds in rainforest and in oil palm plantations that replaced some of the forest. They walked along transects through each area, recording every bird they saw or heard.

a) Suggest why the scientists used transects rather than quadrats.

b) Give two things the scientists should keep the same so that the results from the two areas can be compared fairly.

c) Oil palm plantations have trees planted in rows, all the same age; rainforest has many kinds of trees of different sizes. Predict in which area the birds would show a more regular distribution, and give a reason.

C2. There are many types of grass, each with slightly different leaf shapes. Describe how you would find out which of four species was the most abundant in a large field.

My thinking (any language)

During an outbreak, scientists estimate the number of infected people from samples. Suggest one reason why the estimate might be too low, and one reason why it might be too high.

Word bank abundance · distribution · sample · estimate · quadrat · transect · pitfall trap · sweep net · random · clumped · biodiversity

✂ ---------- cut here

Exit ticket 9C Lesson 6: Ecology: abundance and distribution Name _____

1. Why do ecologists take samples instead of counting every organism?

2. 5 daisies were found in one 0.25 m2 quadrat. The lawn is 40 m2. Estimate the daisy population.

3. Give one reason why quadrats should be placed at random.

Today I can: Goal 1 ☐ Goal 2 ☐